Quantum Mechanics I — Exercise Sheet 9

Position and momentum on the ring

Variant: Questions only

Conventions, as in Lecture 9: $\zeta = e^{2\pi \mathrm{i}/N}$; $(F\psi)_k = N^{-1/2}\sum_x \zeta^{-kx}\psi_x$; the clock $De_x = \zeta^x e_x$, the shift $Te_x = e_{x+1}$ and the mirror $Me_x = e_{-x}$; the site and mode probabilities $p_x = |\psi_x|^2$ and $q_k = |(F\psi)_k|^2$.

Exercise 1 (the square of the transform). Show that $F^2 = M$ and $F^4 = \mathbb{1}$, and conclude that every eigenvalue of F is one of 1, -1, i, $-\mathrm{i}$.

Exercise 2 (kicks and translations). For a state ψ of the ring, compare the two distributions of $D\psi$ and of $T\psi$ with those of ψ.

Exercise 3 (Weyl's uniqueness). Let D' and T' be unitaries on $\mathbb{C}^N$ with $D'T' = \zeta T'D'$ and $D'^N = T'^N = \mathbb{1}$. Show that there is a unitary W with $WDW^{-1} = D'$ and $WTW^{-1} = T'$.

Exercise 4 (equality in the support bound). Take $N = 6$. For each of the states

$$e_0, \qquad \tfrac{1}{\sqrt{2}}(e_0 + e_3), \qquad \tfrac{1}{\sqrt{3}}(e_0 + e_2 + e_4), \qquad \tfrac{1}{\sqrt{6}}\textstyle\sum_x e_x,$$

find the support of $F\psi$ and check that $|\mathrm{supp}\,\psi| \cdot |\mathrm{supp}\,F\psi| = 6$. Then show that $\frac{1}{\sqrt{2}}(e_0 + e_1)$ does not reach the bound, and that $\frac{1}{\sqrt{2}}(e_1 + e_4)$ does.

Exercise 5 (a qutrit's four bases). Take $N = 3$, so $\zeta = e^{2\pi \mathrm{i}/3}$ and $1 + \zeta + \zeta^2 = 0$.

1. Show that $(DT)^3 = \mathbb{1}$, and find an orthonormal eigenbasis of DT.
2. Show that this basis is mutually unbiased with the site basis and with the mode basis f_k: every overlap has squared modulus $\frac{1}{3}$. Show that the eigenbasis of D^2T consists of the complex conjugates of these vectors, and that it is mutually unbiased with the other three bases.
3. For $\rho = |w\rangle\langle w|$ with $w = \frac{1}{\sqrt{2}}(e_0 + e_1)$, find the nine coefficients $c_{rs} = \mathrm{Tr}\big(\rho (D^r T^s)^\dagger\big)$ and check that $\rho = \frac{1}{3}\sum_{r,s} c_{rs} D^r T^s$.
4. Find the probabilities of the reading of w in the eigenbasis of DT, and recover c_{11} from them.

Exercise 6 (⋆ position and momentum matrices). Let $X = \mathrm{diag}(0, 1, \ldots, N-1)$ in the site basis and $K = F^\dagger X F$.

1. Show that $e^{2\pi \mathrm{i} X/N} = D$ and $e^{-2\pi \mathrm{i} K/N} = T$.
2. Show that every diagonal entry of $XK - KX$ is 0, and that $XK - KX \neq 0$. Why does this rule out $XK - KX = \mathrm{i}\hbar' \mathbb{1}$ for any number $\hbar' \neq 0$, without the trace argument?